GEK-8123

DRIVE SYSTEM INSTRUCTIONS

IC4961 TCR D-C DYNAMOMETER CONTROL

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

IC4961 TCR D-C DYNAMOMETER CONTROL

DOM II

The information herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, Drive Systems Department, USA.

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DESCRIPTION

General

The DC dynamometer system is an adjustable voltage system using the DC dynamometer as a conventional DC shunt motor.

A motor-generator set consisting of a squirrel cage induction motor or a synchronous motor and a DC generator are used in conjunction with the dynamometer as shown in Figure 1.

When the dynamometer is motoring, power is taken from the AC line and supplied to the dynamometer through the motor-generator set. When the dynamometer is absorbing, power is returned to the AC line.

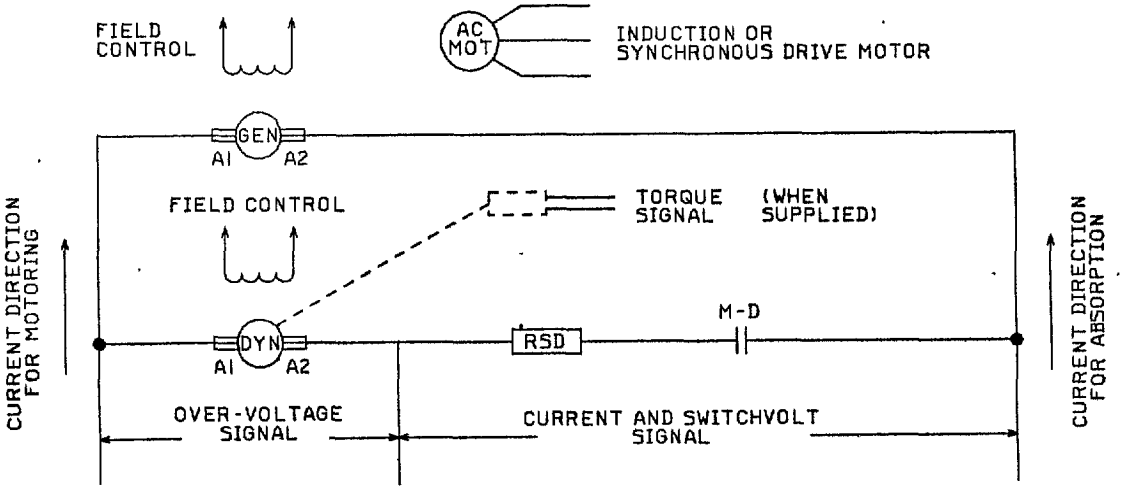


Figure 1

Control of the dynamometer and generator is accomplished by field control of both machines, using an SCR exciter for each. (See below for field adjustment of preamplifier). A closely regulated power supply is used to obtain reference signals, and preamplifiers are used for both the generator and dynamometer controlled by use of suitable input circuits. A block diagram is shown in Figure 2. See the regulator elementary for a system on-line schematic showing adjustment components.

The dashed lines in the block diagram are for torque regulation when supplied.

Adjustment

The proper procedure for adjusting the power supply preamplifier bias, current limit circuits, stability circuits, maximum speed, dynamometer minimum field, dynamometer field current feedback, and overspeed relay are described later.

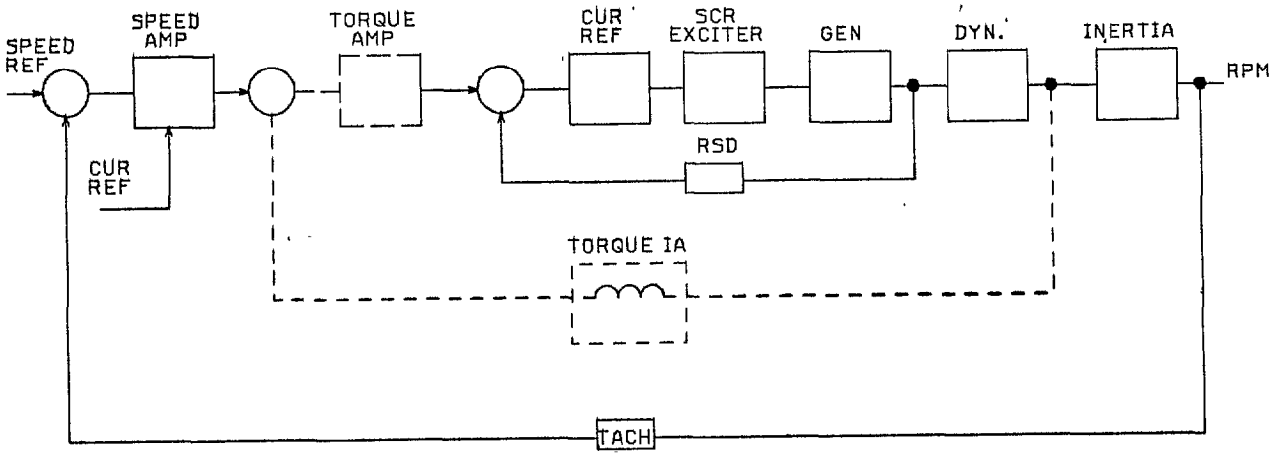


Figure 2
Block Diagram

Power Supply

A regulated power supply is furnished in order to supply constant reference signals - minimizing the drift of the dynamometer operation. See Power Supply specification for details. Specification numbers are shown in the elementary diagram.

Speed Control

The system is capable of closely maintaining speed with a 100 percent change in load torque. A DC tachometer-generator, coupled to the dynamometer, supplies a voltage directly proportional to speed. This voltage is compared with a voltage reference set by a speed adjustment potentiometer on the operator's panel which is connected to the regulated reference supply as shown in Figure 3.

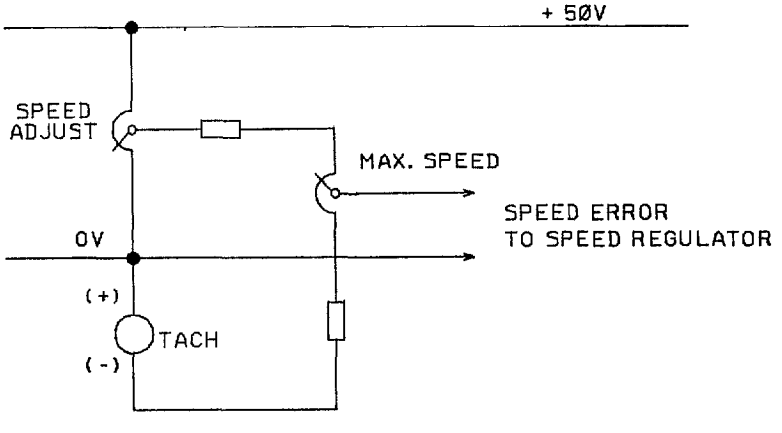


Figure 3
Speed Error Circuit

If the speed signal does not agree with the reference, the resultant error voltage will be amplified by the preamplifier (See Figure 4) which supplies a corrective signal to the current amplifier.

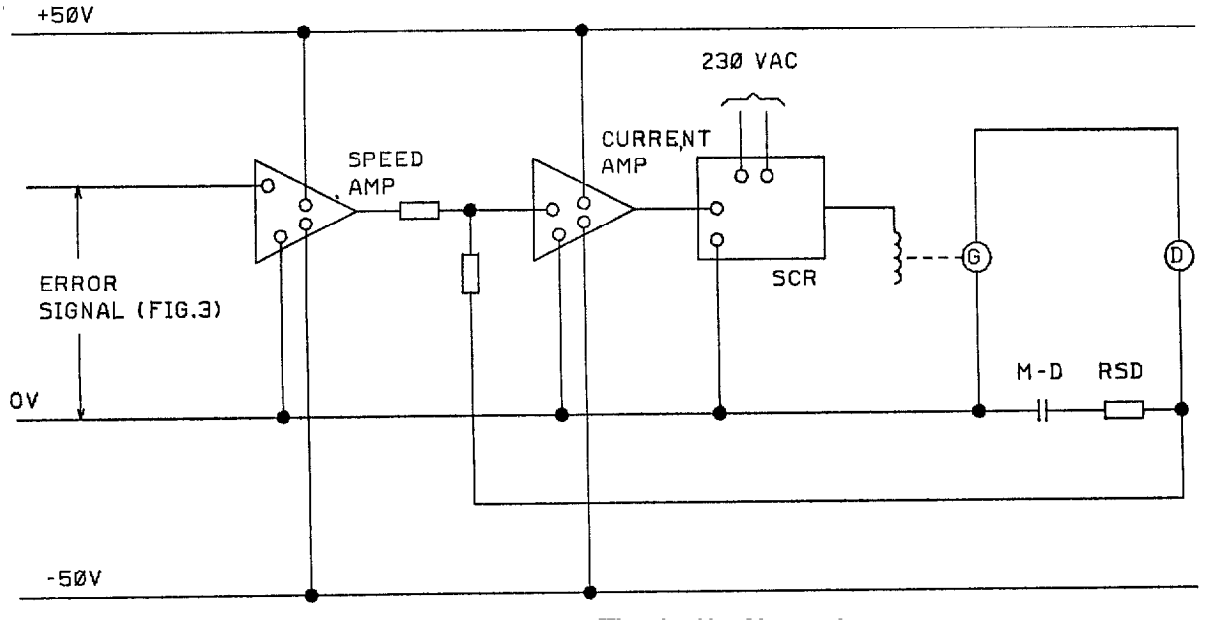


Figure 4

The SCR which is described in the SCR manual excites the generator field. The generator changes the speed of the dynamometer (by changing its terminal voltage) in the direction required to rebalance the tachometer-generator and reference signals.

Dynamometer Field Control

The dynamometer field current is supplied by the dynamometer silicon controlled rectifier package SCRD. This SCR unit is controlled from a pre-amplifier which serves as the SCRD dynamometer field current regulator. The desired field current is set by the speed rheostat in speed and current control. If torque regulation is supplied and the torque mode is selected, the dynamometer field current is set by the tachometer signal. The voltage drop across the resistor RSF in series with the dynamometer field is compared with this voltage reference. The error signal is amplified by the preamplifier and SCR in order to maintain the desired dynamometer field current.

When a speed higher than base speed is desired, some method of changing the reference to the dynamometer field current regulator is needed. This tapering of the field is accomplished by a shaping circuit. The output from the shaping circuit modifies the reference to the field current regulator, thus changing the field current. See IC3500AFT for operation and adjustment of the field taper (See Figure 6).

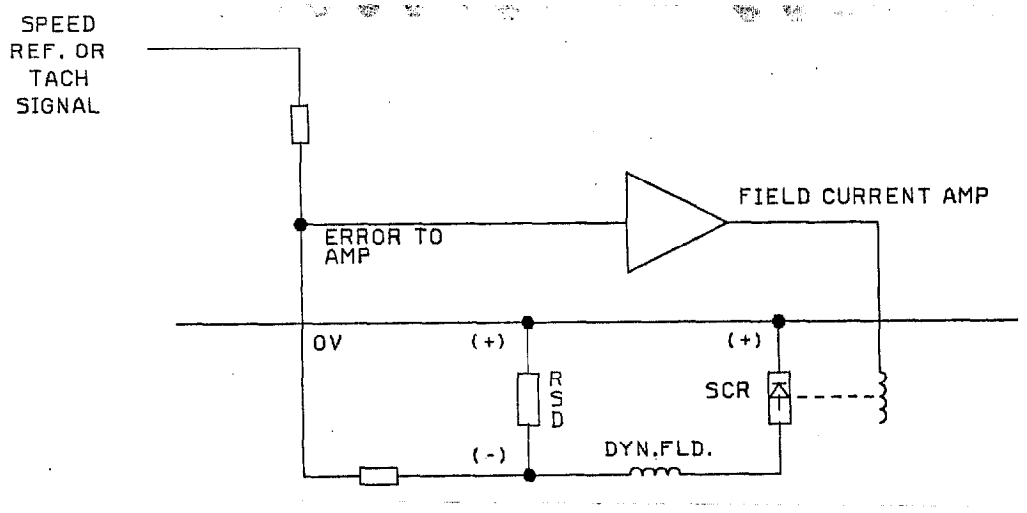


Figure 5

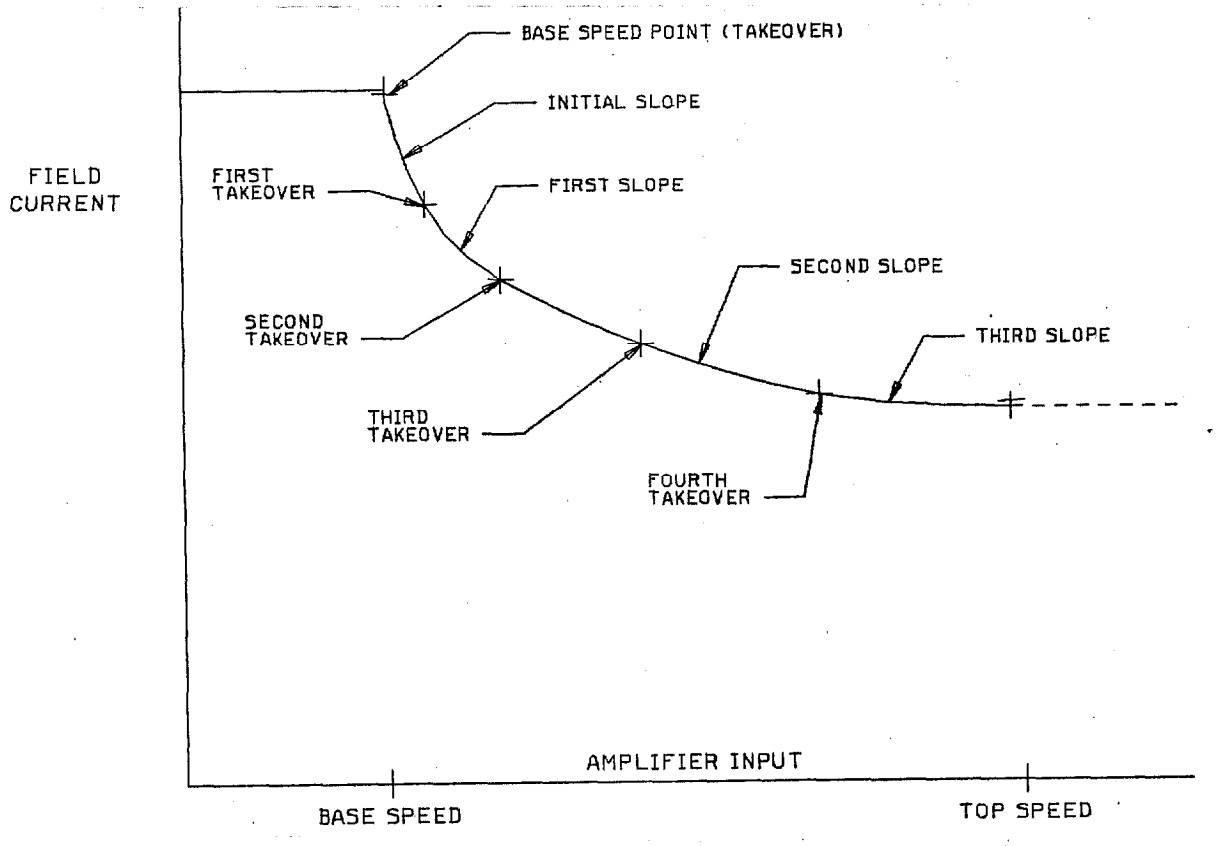
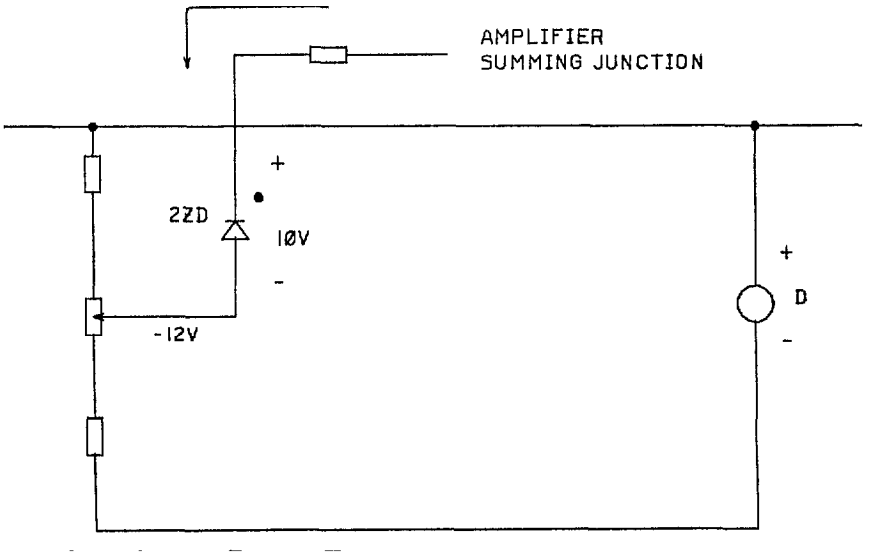


Figure 6
Dynamometer Field Current Characteristics

If the voltage across the dynamometer armature should become too high (for instance, if the test engine should drive the dynamometer above base speed with full dynamometer field current), the voltage limit circuit will operate through Zener diode 2ZD, causing the dynamometer field current to be lowered, as seen in Figure 7.



Current flowing out of the summing junction decreases dynamometer excitation.

Figure 7

For operation of the system above base speed of the dynamometer, in the speed regulation mode, the generator voltage is maintained approximately constant and the dynamometer field current is reduced. This is a constant horsepower type of operation, in contrast to the constant torque operation below base speed. As the dynamometer field current is lowered by the speed control potentiometer on the operator's panel when operating above base speed, the dynamometer tachometer-generator is still supplying a signal which is compared to the setting of the operator's speed control potentiometer. This error operates as before to adjust the generator field, hence dynamometer armature voltage. Therefore, regulation of speed is still accomplished by means of dynamometer armature voltage. Above base speed, the armature voltage is primarily a trimming action in that main control is obtained by dynamometer field weakening. The operator must be careful not to exceed the horsepower rating of the system while above base speed.

When operating in the armature current regulation mode, the speed pot is rotated to a speed higher than one that will be reached under current control. This permits operation at all speed levels.

In the torque mode of operation (if present), the field current is programmed from the tach signal. This permits the drive to operate at all speeds at greater torque levels than in the current regulation mode.

Current Limit Circuit

Maintaining speed within close limits requires a high gain system. This means that a small error between the tachometer generator voltage and the reference voltage will cause a large change in motor armature voltage. This large change could cause the armature current to become excessive. Therefore a circuit which limits motor armature current is included to protect the machinery. This is accomplished by limiting the output of the speed preamplifier (see Figure 8). The error necessary to move the speed preamplifier is very small, therefore its potential can be considered as being zero volts at all times.

If the speed error causes the output of the speed preamplifier to exceed the voltage set by either the motoring current limit potentiometer or the absorption current limit potentiometer, the diodes shown in Figure 8 will conduct to limit the preamplifier output to these set values, thus limiting the current. These circuits are adjusted to operate between 125 and 140 percent of rated armature current.

Voltage V_1 is produced by the polarity and magnitude of the speed error. (See operational amplifier techniques applied to regulating systems).

When the voltage of V_1 produces enough current to cause I_2 R_1 to exceed V_1 (point A becomes plus with respect to 0V bus), then current I_1 will flow. V_2 and I_3 are always zero. Enough current will flow to cause I_1 R_i to equal the error voltage. The amplifier output cannot now increase except by the small amount of voltage (I_1 R_1) needed to keep I_1 R_1 equal to the error.

Current Control

A control switch on the operator's panel can be used to select speed control or current control of the system. When under current control, the dynamometer field current will still be regulated by the speed control potentiometer. The field current (speed potentiometer) should be adjusted to a value corresponding to a speed higher than that which will be reached under current control. The current control potentiometer can then be used for regulating current output or input of the dynamometer system in a manner similar to the speed control operation.

The speed preamplifier is used as a summing amplifier whose sole function is to assemble signals from their respective inputs and pass them through the current limit circuit into the current preamplifier (See Figure 9). Operational amplifiers have the characteristic shown in Figure 11 when used for summing signals.

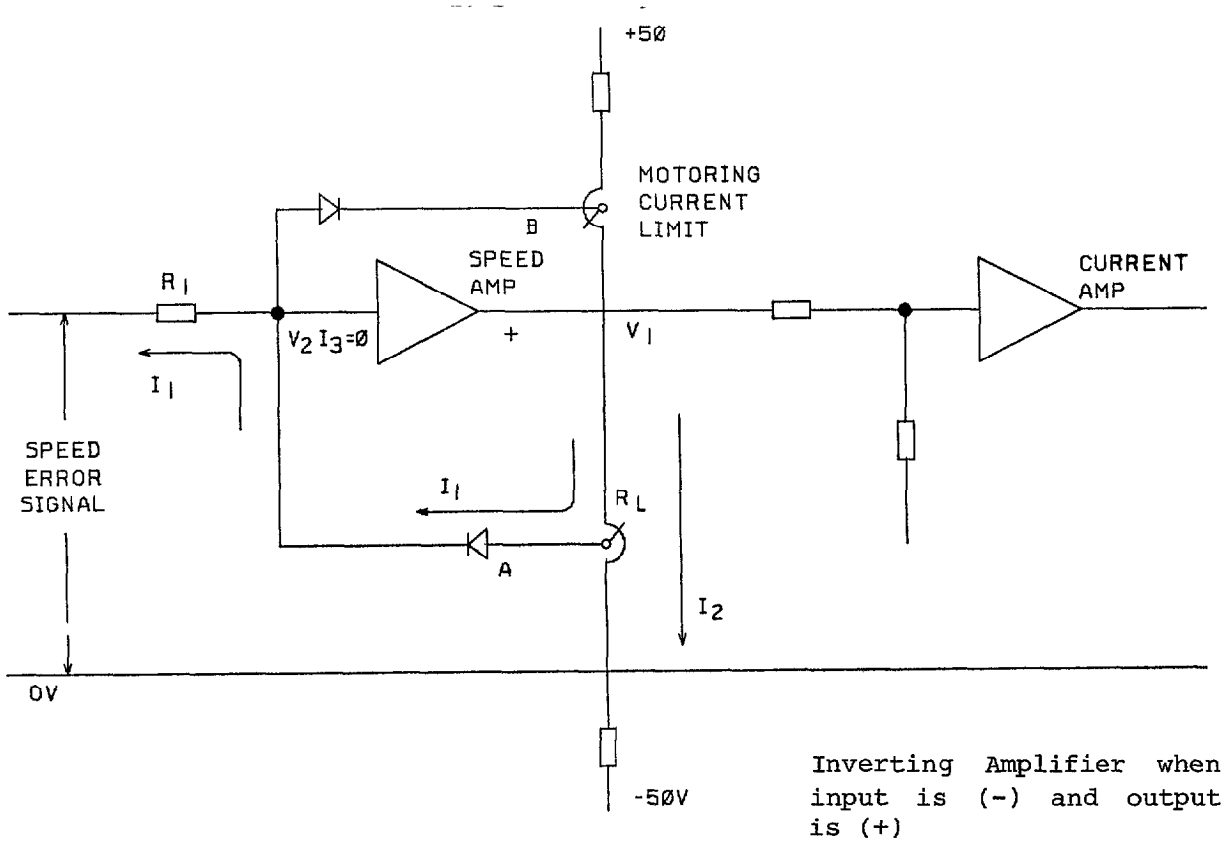


Figure 8
Current Limit Circuit

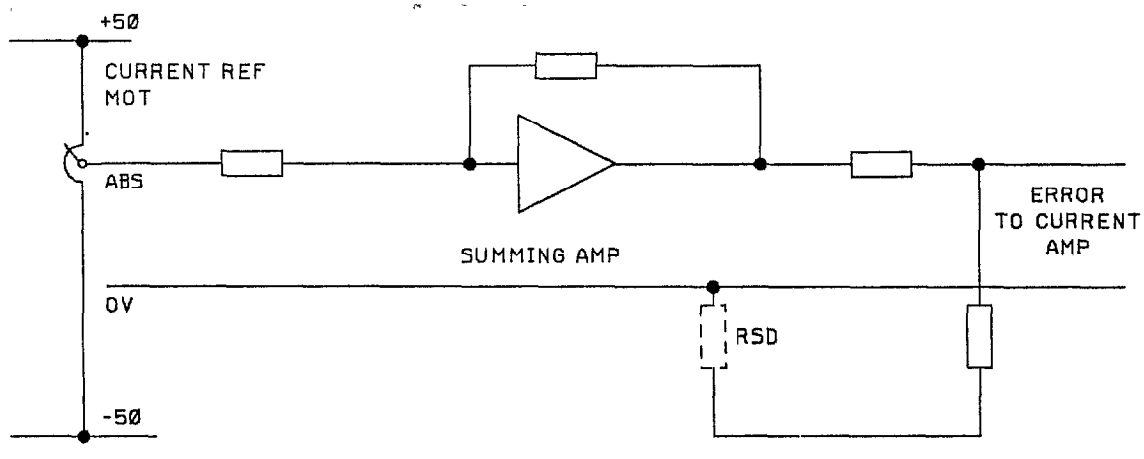


Figure 9

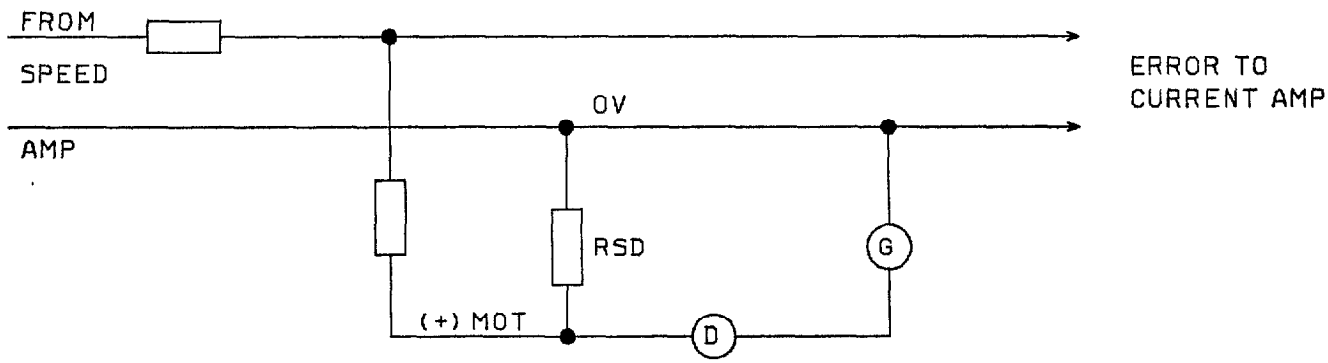


Figure 10

A signal from the speed preamplifier is compared to a signal obtained from a resistor in series with the dynamometer and generator (See Figure 10). Any error is amplified by the current preamplifier and then applied to the SCR exciter. The SCR exciter changes the generator voltage as necessary to hold the current error to a small value. See Figure 4 for current amplifier.

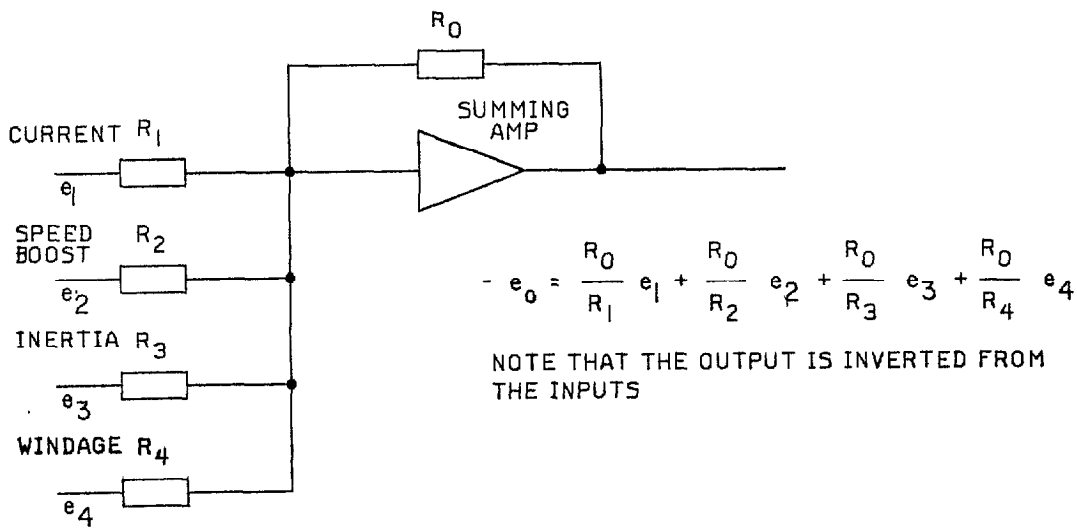


Figure 11
Summing Amplifier

The output of the summing amplifier is the current reference signal.

This error voltage then goes through the same preamplifier, SCR exciter, and generator to supply armature voltage to the dynamometer. The operator's current control potentiometer will supply either a positive or negative reference voltage to obtain constant motoring or absorption current. Since the voltage drop across the resistor at rated current is only about 7.5 volts, the current regulator is not capable of as close a regulation as the speed control system. As before, the current limit circuit will still operate in a manner necessary to protect the armature curcuit against overcurrent.

Speed Boost

The operator's panel also contains a speed boost potentiometer (See Figure 12). Making adjustment here allows the operator to add a signal proportional to speed in the input circuit while under current control (torque control if available). This will provide a rising absorption-torque characteristic as the speed is increased. In the case of an engine which has a flat region in its speed-torque characteristic, this speed boost signal will allow stable operation of the dynamometer system and the engine.

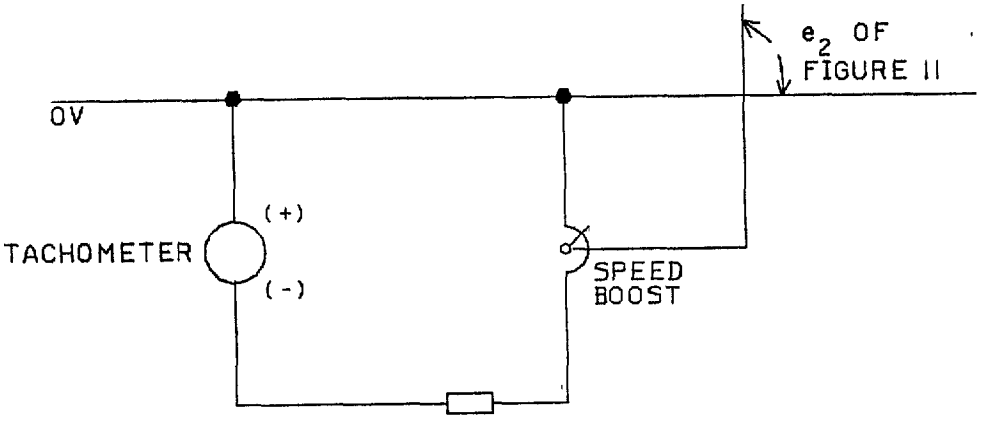


Figure 12

Windage

The windage function, produces a change in current (torque control if available) proportional to the square of speed, is also included in many drives. (See Figure 13). This circuit simulates the windage force that is exerted on a vehicle.

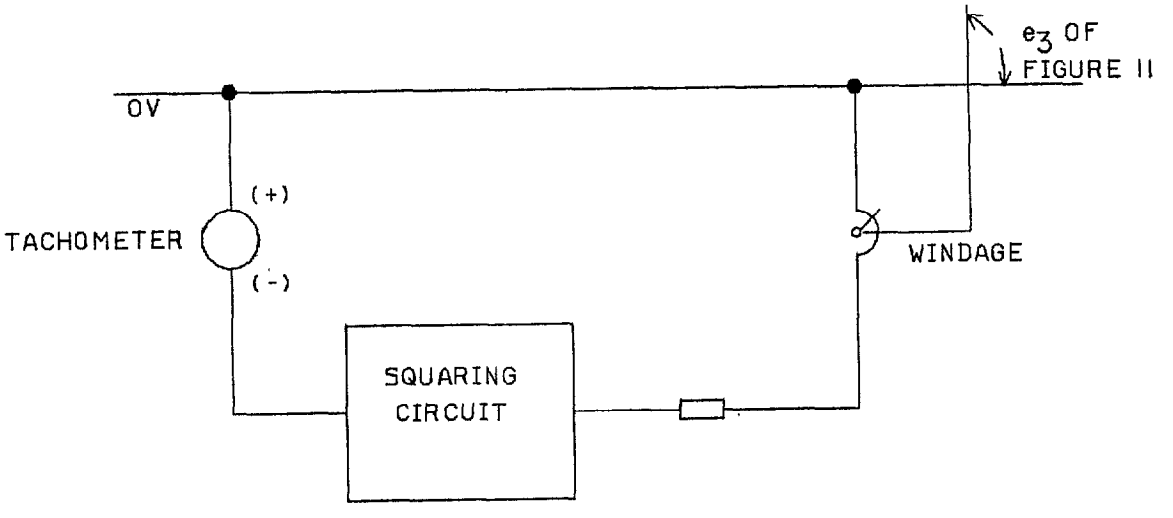


Figure 13

This function is used when road load simulation is being used on either engine or chassis dynamometers.

Electrically Simulated Inertia

To electrically simulate an inertia load on a test motor, a load must be applied which is proportional to the rate of change of speed. This is done by operating in the current or torque mode and adding to the reference an absorption reference proportional to the rate of change of speed. This simulates a positive inertia. An added reference of the opposite polarity simulates negative inertia. Inertia is simulated with the following circuit:

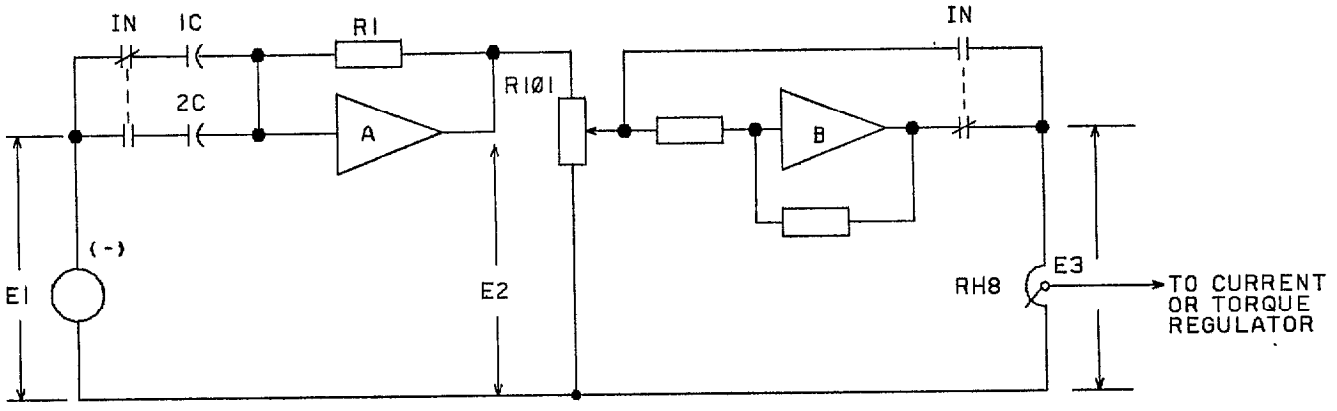


Figure 14

If the inertia selector switch is in the POSITIVE position, the IN relay is de-energized and the IN interlocks are as shown above. If the dynamometer is accelerating, a current will flow in 1C as shown below:

$$I = \frac{\Delta E_1}{\Delta t} \quad (1C), \qquad E_2 = IR_1 \frac{\Delta E_1}{\Delta t} \quad (1C)\ (R_1)$$

If the acceleration rate is constant then E_2 and E_3 are constant. A constant reference is supplied to the regulator. The maximum reference may be adjusted with R101 when RH8 is fully clockwise.

For negative inertia, "B" amplifier is bypassed, therefore, E_3 is of opposite polarity. The inertia signal is proportional to a current flowing through 2C.

Torque Regulation

If torque regulation is furnished, a control switch on the operator's panel can be used to select the speed mode or torque mode. In the torque mode the speed reference and tach signal is shorted to common and the desired torque reference is set by the torque adjustment pot. This reference is compared with a torque signal which is usually obtained from a strain gauge on the dynamometer. This is a low level signal (millivolt range) and must be amplified to the regulator voltage level. This requires a high gain amplifier.

Any noise pickup by the torque feedback circuit would be amplified by the feedback amplifier and cause poor regulator performance. For this reason, an isolation amplifier is used.

Any error between the torque reference and the feedback is amplified by the torque and current preamplifiers and then applied to the SCR exciter. The SCR exciter changes the generator field current as necessary to hold the error to a small value.

The speed preamplifier is used as a summing amplifier whose sole function is to assemble signals and pass them through the torque limit circuit into the torque preamplifier (See Figure 15).

The torque regulator is similar to the speed regulator in that it has an inner armature current loop. The output of the torque amplifier is the armature current reference.

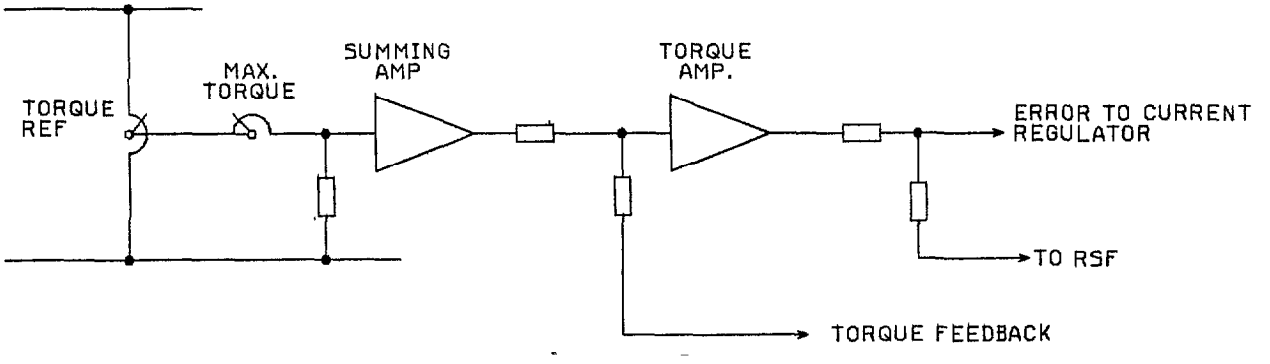


Figure 15

As in the case of current control, if windage, speed boost and inertia are available, they add to the torque reference and are summed with the speed amplifier. See Figure 11.

Torque Limit

In torque control (if present), the current limit pots are around the torque amplifier (See Figure 15). It operates as explained earlier in these instructions. The torque limit circuit limits the output of the speed amplifier and also operates similar to the current limit circuit explained earlier.

Emergency Stop

If the EMERGENCY STOP pushbutton is pushed, the control will automatically be transferred to speed control with zero speed reference. This will cause the regulator to operate from a speed signal modified only by current limit. The dynamometer will thereby be stopped rapidly by regenerative braking. User's protective devices, such as bearing temperature relays, cooling water pressure switches, etc., can be connected in to give an emergency stop. An ignition relay will also be dropped out, and will interrupt ignition for the test engine. During regenerative braking, the current limit circuit prevents a loop circuit overcurrent.

Switch - Volt Relay

The switch-volt relay is connected across the loop circuit contactor, and will prevent closing this contactor as the voltage across its tips exceeds approximately 60 volts.

Switch - Volt Regulation

Switch volts refer to the voltage across the loop contactor when it is de-energized. Switch-volt regulation is the term applied to the function of regulating switch volts to a low value (within 5 percent of rated voltage) when the loop contactor is de-energized whether or not the dynamometer is rotating.

When the loop circuit contactor is opened, the voltage across the contactor is fed into the current preamplifier. This signal causes the current preamplifier to maintain switch-volts at approximately zero.

Speed Error Detecting Relay

The speed error detecting relay compares the actual speed of the dynamometer as measured by the tachometer-generator with the desired speed as set by the speed control potentiometer; if this difference exceeds approximately 1.5 volts, a relay will be energized and will prevent closing of the master relay, MR. This prevents the operator from attempting to reclose the loop circuit contactor in case the indicated speed, as set by the speed control potentiometer, is not consistent with the actual dynamometer speed.

Operational Amplifiers

It should be recognized that the internal circuitry of the operational amplifiers is not adapted to field repair. Therefore, it is recommended that no repair be attempted on the amplifier. If it is determined that the amplifier is defective, remove it and plug in a different amplifier. Return the defective amplifier to the factory for repairs.

Silicon Controlled Rectifier Exciter

This unit has been tested in the factory and has been adjusted for operation with the machines listed on the data sheet of the elementary diagram. If trouble develops or if a check is desired, follow the procedure given in the SCR manual.

Stabilizing Circuit

The system must have a high gain in order to obtain the accuracy desired. In high-gain systems, where a feedback is compared with a reference signal, instability may become a problem.

Instability is caused by energy storage devices and components within the closed loop system - in particular the inertia of the dynamometer and test engines, but also the armature loop circuit inductance, generator field inductance, and amplidyne inductances.

Stabilization networks are shown later.

Protective Features

Included in the control system are various protective devices in addition to the EMERGENCY STOP pushbutton discussed above. Only the following items will open the loop circuit contactor:

1. Motor-generator set overspeed (centrifuge switch on motor-generator set).
2. Dynamometer overspeed (voltage sensitive relay operating from tachometer-generator).
3. Armature loop circuit instantaneous overcurrent relay.
4. Dynamometer field loss relay.
5. STOP pushbutton.
6. Dynamometer overvoltage relay.
7. Auxiliary starters or main motor-generator set starter.

As mentioned above, the speed error detecting relay and the switch-volt relay will prevent the operator from closing the line contactor unless the voltage across it is small.

OPERATINAL TECHNIQUES APPLIED TO REGULATING SYSTEMS

The term operation imposes several conditions on any amplifier it is applied to. These are:

1. Gain is sufficiently large that input voltage and current are considered to be zero.

2. There is no internal time constant, i.e., bandwidth on frequency is very wide and rise time on step input from 5% to 95% of output is negligibly small.

The above conditions allow the following conclusions. For the configuration shown,

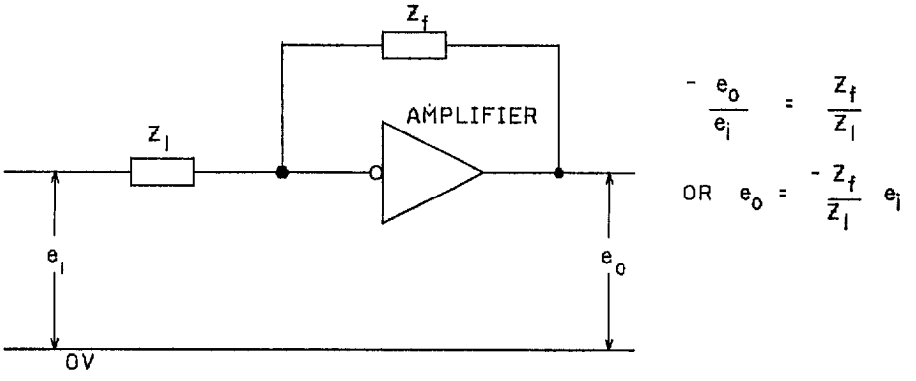


Figure 16

Z_i and Z_f can be any combination of capacitance, inductance, and resistance. The negative sign indicates 180° phase shift (polarity change) from input to output.

The following elementary circuits are commonly used.

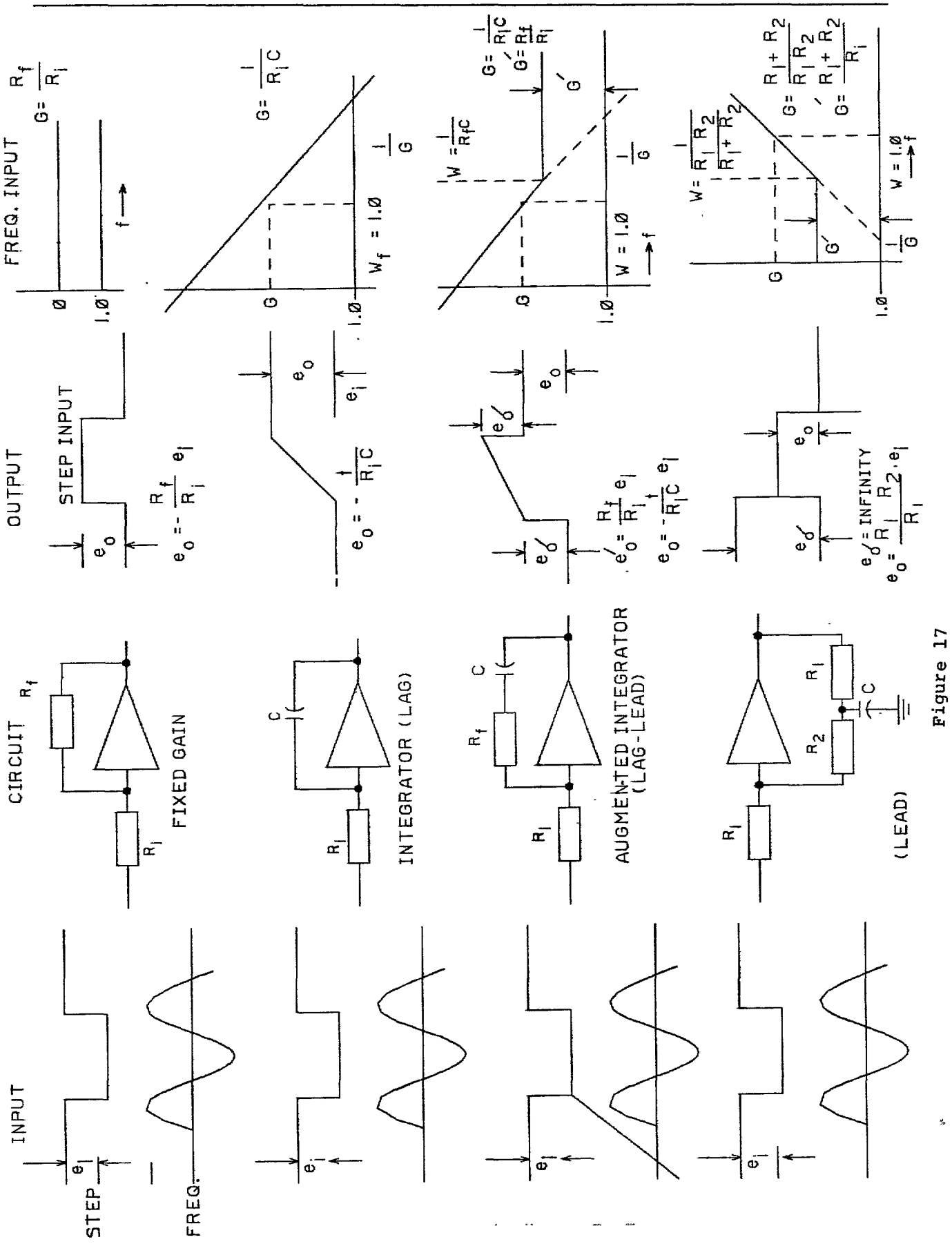

CIRCUIT
FIXED GAIN
INTEGRATOR (LAG)
AUGMENTED INTEGRATOR (LAG-LEAD)
(LEAD)
OUTPUT
STEP INPUT
FREQ. INPUT
INPUT
STEP
FREQ.

Figure 17

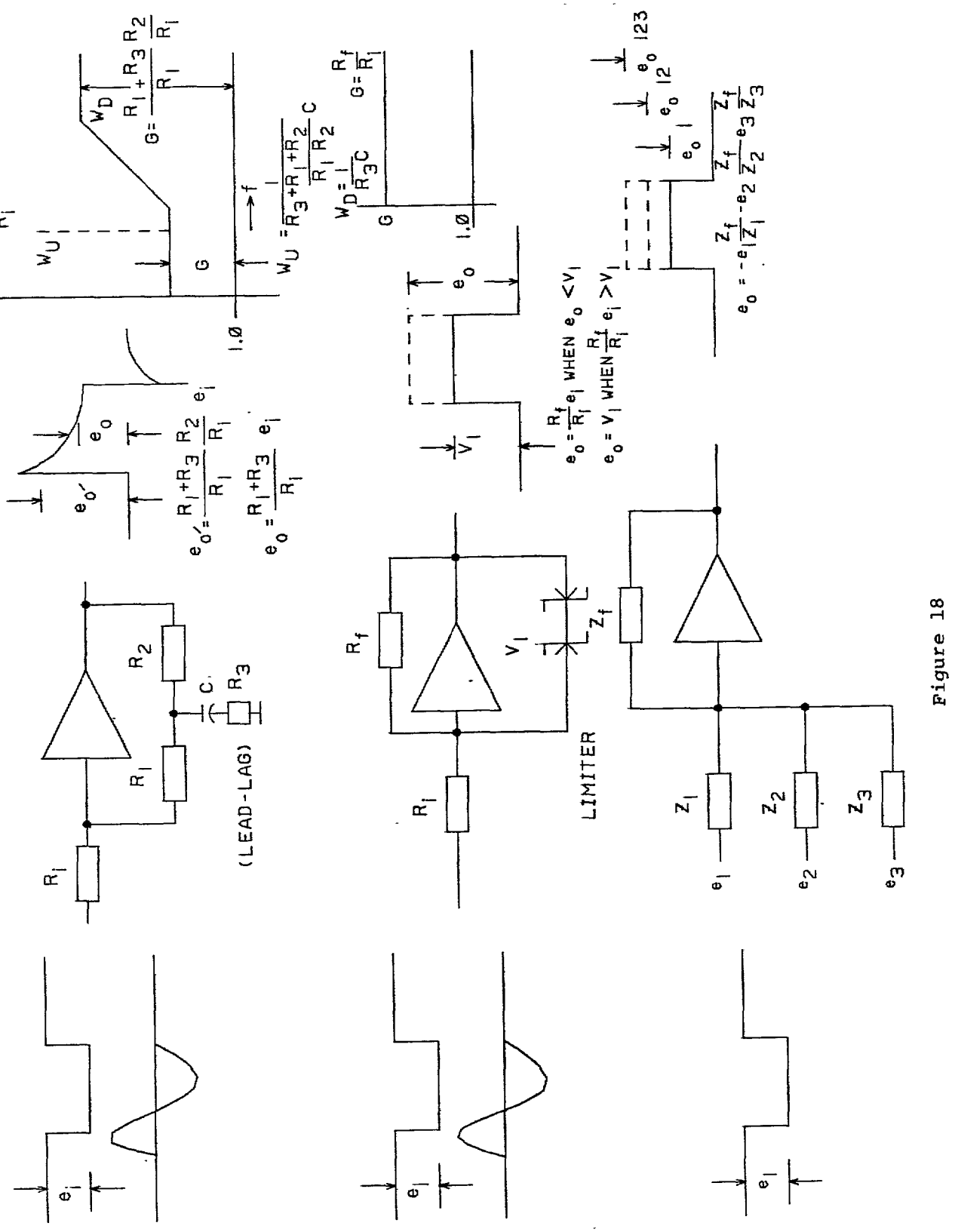


Figure 18

Regulating System Notes

In any regulated system using negative feedback, the phase shift from input to output must be less than 180°. Any device which has capacitance or inductance causes phase shift and attenuation. The corner frequency at which attenuation begins is the reciprocal of the time constant. In a generator field, attenuation would begin at:

$$\frac{R_f}{L_f} = \frac{1 \text{ RADIANS PER SECOND.}}{T_f}$$

If more than one time constant exists in a regulated system, stabilizing means must be added. Stabilizing normally consists of increasing the gain of the regulator in proportion to the decrease in gain (increase in attenuation) of the other system elements. Then additional controlled attenuation is added to the regulator to control the speed of response of the system.

In general (rule of thumb), response in radians per second is equal to $\frac{3}{t}$ for the following condition. (See Figure 19)

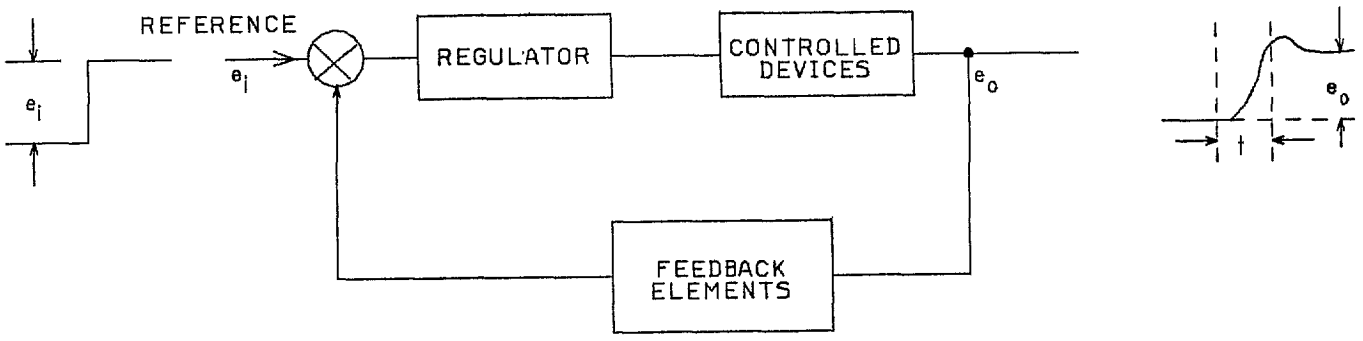


Figure 19

For the general system, any time constant lower than the response frequency or less than 3 times greater than the response frequency must be compensated. The following simple example will illustrate. (See Figure 20).

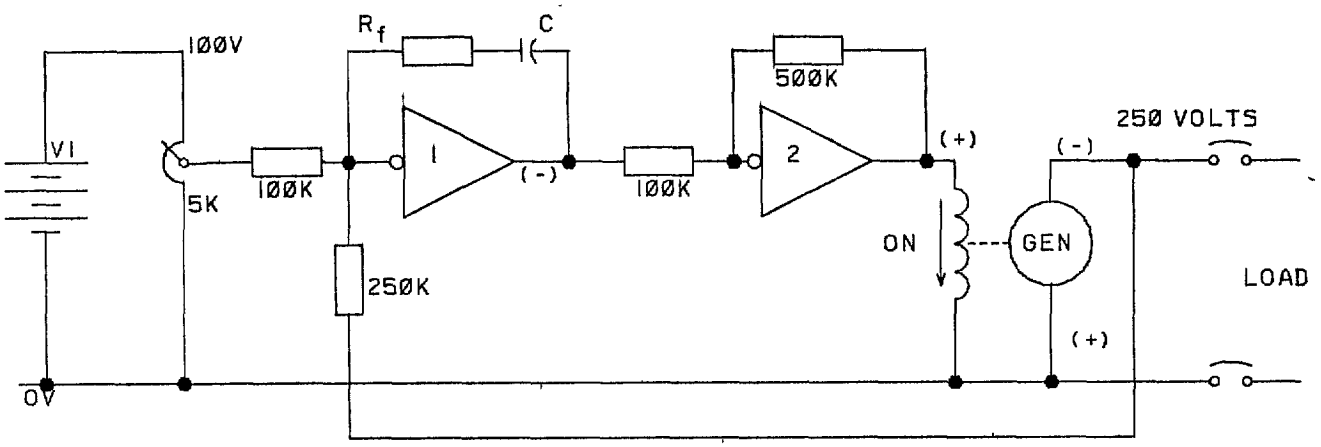


Figure 20

Assume the following:

1. No saturation occurs in any element.
2. Generator field time constant = 1 second.
3. Desired response frequency = 10 radians/second.
4. Generator gain = 5 volts/volt; i.e., 1 volt on field produces 5 volts on armature.

The DC accuracy of the above system is theoretically infinite; i.e., regardless of load, the output voltage will not change if the reference is not changed. The following conditions should exist to fit the foregoing statement.

1. $R_fC = 1$ second
2. G = 10 at W = 1.0 radian or W = 10 radians when G = 1.0

Solution: GH exclusive of amplifier 1 $= \frac{500}{100} \times 5 \times \frac{100}{350}$

$$= \frac{2500}{350} = 7.14 \text{ V/V}$$

Therefore G of amplifier 1 $= \frac{10}{7.14} = 1.4$ V/V at W = 1.0

$$G = \frac{1}{R_iC} \qquad R_i = \frac{100 \times 250}{350} = 71.4 \text{ K ohms}$$

$$C = \frac{1}{R_iG} = \frac{1}{71.4K \times 1.4} = \frac{10^{-3}}{100} = .01 \times 10^{-3} = 10 \text{ MFD}$$

$$R_f = \frac{1}{10 \text{ MFD}} = 100K \text{ ohms}$$

If power amplifier 2 contained a time constant longer than 33 milliseconds, the system should appear as shown below:

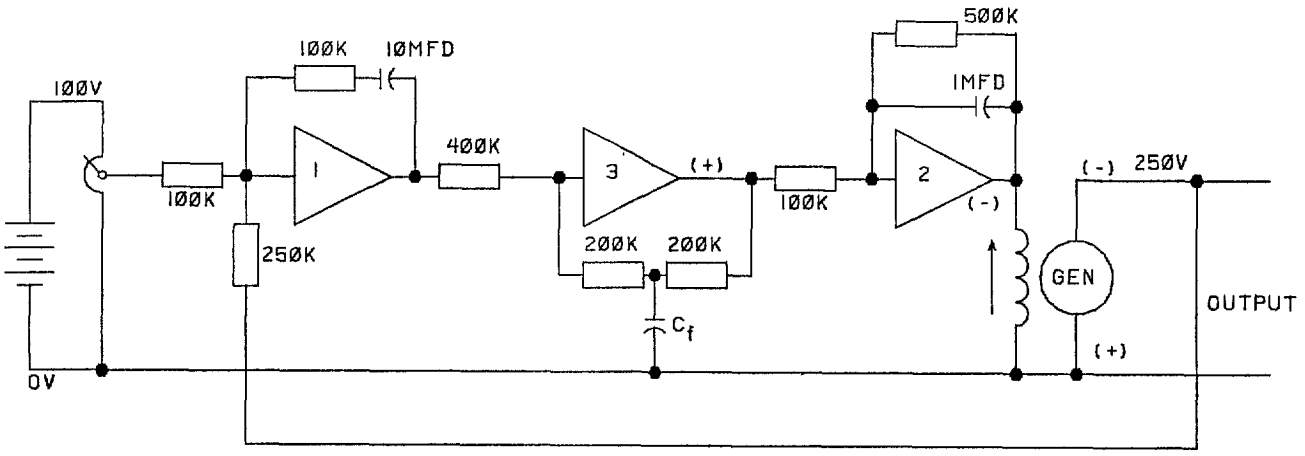


Figure 21

Time constant of amplifier 2 = 500 K X 1 MFD = 1/2 second.

Therefore, the lead time constant of amplifier 3 should equal 1/2 second. Gain should remain constant at DC level.

$$\text{GAIN} = \frac{200K + 200K}{400K} = 1$$

$$W^u = \frac{1}{\frac{R1\ R2}{R1 + R2}} \quad C_f \quad \frac{1}{100K} \quad C_f$$

$$C_f = \frac{1}{100K \text{ X } 2} = 5 \text{ MFD}$$

} Amplifier 3

Transient Response

Response for a perfectly compensated sytem as shown above should appear as shown below.

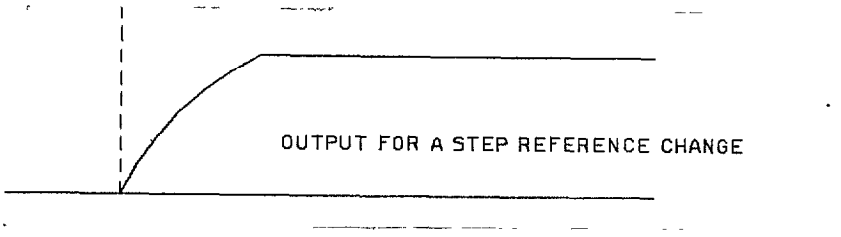
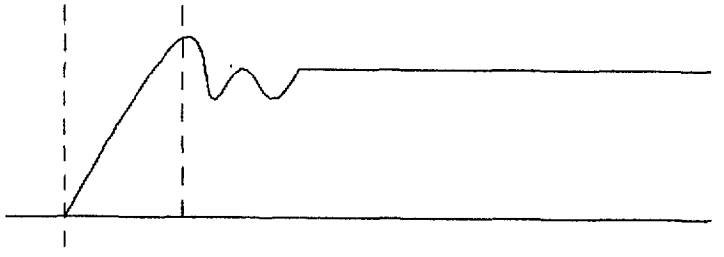


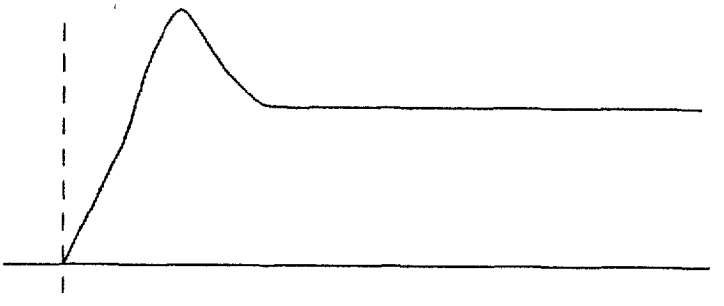
Figure 22

In a system where perfect compensation is not possible (because of changing time constants) the following guides are helpful.

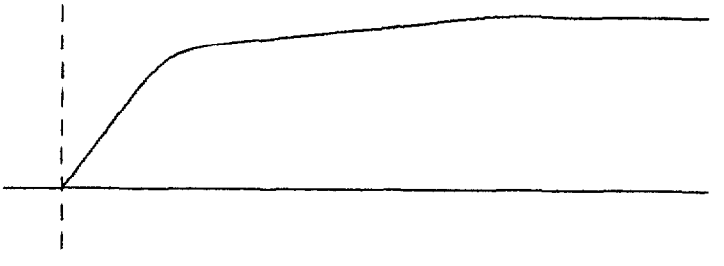


DECREASE RESPONSE (INCREASE C ON AMPLIFIER I)

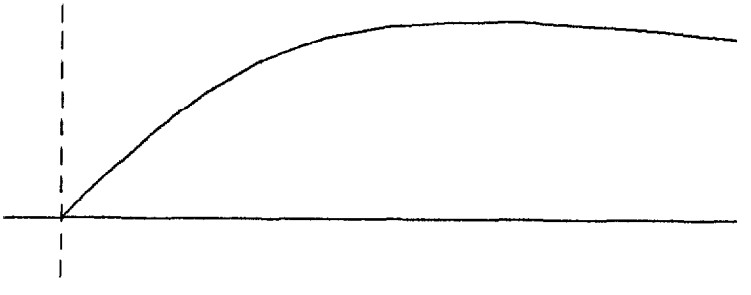
NOTE THAT R_f MUST DECREASE TO MAINTAIN R_fC CONSTANT



INCREASE LEAD (INCREASE R_f OF AMPLIFIER I)



DECREASE LEAD (DECREASE R_f ON AMPLIFIER I)



INCREASED RESPONSE (DECREASE C ON AMPLIFIER I)

NOTE: INCREASE R_f TO MAINTAIN R_fC CONSTANT

Figure 23

Saturation of any element will affect the above responses.

Internal Loop Systems

The foregoing discussion has taken into account only one closed loop. Discussion of systems having internal loops or parallel regulators is too complex to be given here. In general, the response of any internal loop should be 3 times faster (and preferably 5 to 10 times faster) than the next outer loop. (See Figure 24).

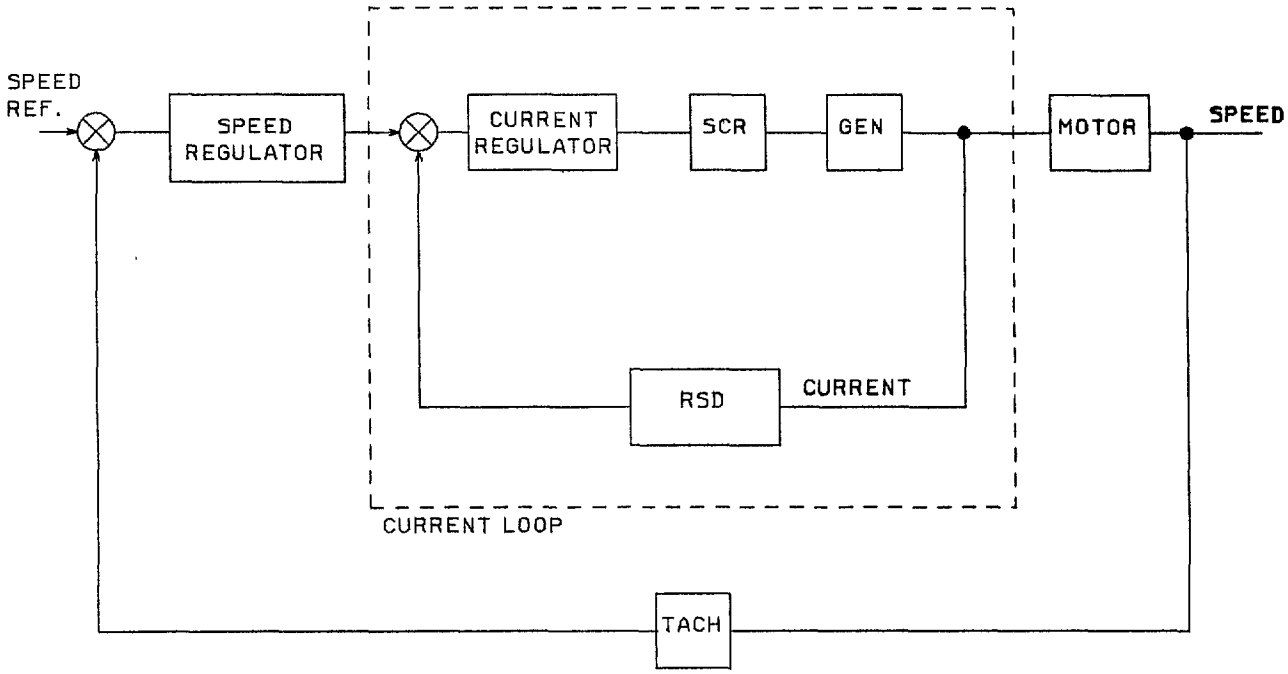


Figure 24
Current Regulator

The current loop should be faster than the speed loop. If the current regulator is always to control the machine current, systems having motor field regulators in addition to the speed and current regulators shown in Figure 24 will need special attention. The motor field should always change at a rate equal to the rate of speed change. Systems requiring very fast accelerating or decelerating speeds or requiring response to step reference changes need counter EMF or terminal voltage regulators on the motor.

INSTALLATION AND ADJUSTMENTS

Receiving and Unpacking

The control equipment is mounted in a rigid floor-mounted enclosure, and panels are crated and shipped in an upright position. When a panel is received, it should be handled from the top and kept upright.

A panel which is not to be installed immediately should be stored in a clean, dry location. When the panel is installed, the control should be unpacked carefully, making certain that no small parts are thrown away with the packing material.

Check the nameplate rating of the machine to be sure that it agrees with that given on the panel nameplate. The control is designed for use with specific equipment and must not be applied to any other without first checking with the nearest Apparatus Sales Office of General Electric Company.

After the panel has been unpacked, it should be handled from the removable lifting angle on top provided for that purpose. To avoid undue strains on the panel, sling the lifing rope through the holes near the ends of the lifing angle.

Clean the inside of the control panel with a brush, soft cloth, or dry compressed air. All dirt, dust, or bits of packing material, which interfere with successful operation of the panel devices, should by all means be removed from the panel. You should likewise take care during the cleaning operation to prevent any dirt from being blown into the inaccessible spaces of the devices.

Installing the Panel

When the panel is to be installed, it is essential that it be securely fastened in a true, upright position on a level surface to permit proper operation of the devices mounted on the panel.

The enclosing case is mounted on two floor sills, one each across the back and front of the enclosing case. Floor sills consist of channel irons which are four inches deep and one and three-quarters inches high. The four-inch dimension is horizontal. On single-case installations the floor sills may be removed and the enclosing case bolted on the floor. On lineups of more than one case it is recommended that floor sills be used to give rigidity to the panel. Mounting holes for one-half-inch bolts are provided in each sill. The outline for the particular panel to be installed should be checked to determine the location of the mounting holes. It is recommended that the one-half-inch mounting bolts be grouted in the floor in the proper location for fastening the panel.

There are several possible methods of installing the floor sills. The most frequently used methods are described below:

If the panel is to be installed on an existing floor, grout the one-half-inch anchor bolts in the floor and slip the panel over them. Check level position of the sills with a machinist's level, and shim the sills if necessary. Check that the surfaces of the front and back sills are in a plane, so the angle iron supports of the panel are plumb. If the panel is installed in this manner, there will be a two-inch gap between the floor and sides of the panel enclosure. This gap should be closed to prevent accumulation of dirt under the panel, which in due time may interfere with the proper operation of the panel devices.

If the concrete floor is to be poured at the time that the panel is installed, grout the one-half-inch anchor bolts in the sub-floor, and slip the panel over them. Check alignment of the floor sills with a level and make sure that the panel supports are plumb. Then pour concrete around the floor sills. This should be done carefully, so as not to splash concrete on the panel devices. Care should also be taken to finish the surface of the concrete so there will be no interference of the enclosing case door and so there will be a tight joint between the floor and the enclosure side, thus preventing the entrance of dirt at the bottom of the panel.

If the concrete floor is to be poured before the panel is installed, a dummy set of floor sills can be used. These dummy sills can be made of wood of the proper size and length. The dummies should be set in place, together with the one-half-inch anchor bolts, before the concrete is poured, and then replaced with the actual sills, after the panel is received.

Provisions should be made for incoming and outgoing cable and control wires before the panel is in place. In all cases the outline for the panel being installed should be used to determine the space available, and the panel interconnection diagram should be checked for the number of wires and cables which will be installed. The cables normally enter the bottom of the panels.

Preparing the Panel for Operation

Before you can operate the panel, even for a tryout without power, all devices must be placed in fully operative condition. The armatures of some of the relays and contactors may have been tied or blocked to prevent damage during shipment. Remove all of these ties and blocks before operating devices.

Scaling surfaces of unplated and laminated AC contactor and relay magnets, as well as the ends of shafts, are coated with a rust preventative. This protection against rust is a heavy grease which does not interfere with the proper operation of the devices, but which encourages the accumulation of dust and dirt which may ultimately prevent proper device operation.

Therefore, before operating the panel, you should wipe off all rust preventative. Magnet sealing surfaces may then be coated with a thin film of light machine oil.

Operate each device and relay by hand to see that the moving parts operate freely without binding. Make sure that all electrical interlocks are clean and make good contact when closed. Relays and contactors are carefully adjusted and tested at the factory. If adjustment information for individual devices is needed, however, it may be found in individual instructions for each device.

With the power leads disconnected at the machine terminals, but with all devices properly readied for operation, apply power and operate the panel to made certain that its operation is in agreement with the elementary diagram.

The leads may then be connected and the equipment put into operation.

Field Taper & Field Current Regulator

The field current regulator and taper has been factory set to calculated values of field current. Some slight amount of field adjustment may be necessary. Dynamometer terminal voltage should stay between 100% plus 10% minus 0% (Example: for a 250 volt machine the voltage should remain between 250 and 275 volts) at any speed after transients have decayed. See the data sheet of the elementary for the end points on the dynamometer taper curve.

The following checks should be made when the dynamometer is not running. When the speed potentiometer is turned very fast, the field current will probably appear as shown. (See Figure 25).

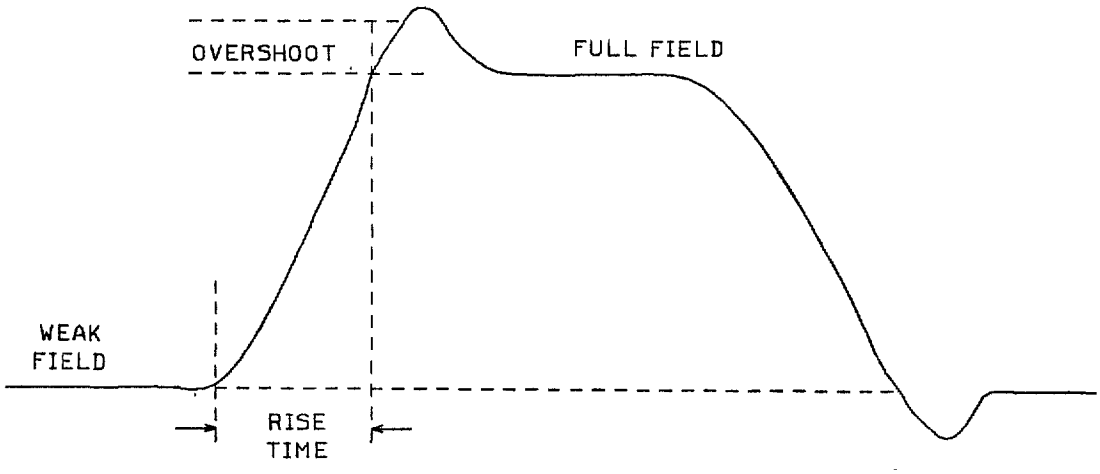


Figure 25

Increasing the ohms in RH104 will reduce the overshoot. Correct adjustment is obtained when the overshoot is less than 5%.

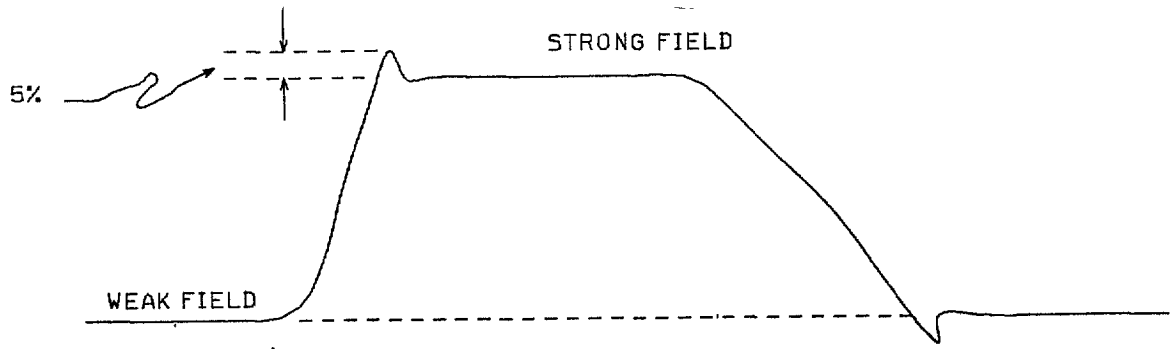


Figure 26

It may be necessary later to adjust RH105 to give faster or slower response for small steps.

If, for some reason, the taper is grossly in error and the entire curve needs to be re-accomplished, proceed as outlined in IC3600AFTA, Sheet 30.

After the taper has been set, adjust the overvoltage takeover. Two methods are given; either may be used.

Method No. 1:

1. Disconnect SV coil.

2. Remove FU4 fuse between ACOM and A2 terminal of dynamometer. Jumper ACOM to A2 of generator.

3. Open coil of MR so that it cannot pick up.

4. Remove speed amplifier.

5. With the bias, raise the voltage of the generator to the value of takeover. This should normally be about 120% of generator volts (120% X 250 volts = 300 volts).

6. Adjust overvoltage takeover on field taper card (AFTB) until SCRD voltage begins to decrease.

7. Lower generator voltage and then again raise it to check takeover. The dynamometer field should weaken quite rapidly when overvoltage is reached.

8. Remove jumper. Reconnect SV and MR coils and reinsert speed amplifier. Reinsert FU4 fuse.

Method No. 2:

Bypass this test until the overspeed relay is checked. Proceed as described therein, except leave RH1 at base speed position. Note voltage of dynamometer as speed increases. Adjust overvoltage takeover to control terminal volts.

Before starting the MG set, check the amplifier as follows:

Check to see that current limit pots RH13 and RH14 (RH4) are turned to 2.0 or higher. Throw the test switch to TEST position.

JR5 should read (-) 5 volts $\pm$ .5 volts.
JR9 should read (+) 5 volts $\pm$ .5 volts.
SCRG should read 50 volts $\pm$ .5 volts.
JR14 should read (+) 5 volts $\pm$ .5 volts (if torque control is present).
JR15 should read (-) 5 volts $\pm$ .5 volts (if torque control is present).

The field current response should not be able to change faster than the maximum rate of change of speed.

If a machine requires 5 seconds to accelerate from base speed to top speed, then field current should change from full field to weak field in 5 seconds or longer. Depending on the generator field hot drops, some difficulty may be experienced in holding current limit on very fast changes in speed reference potentiometer setting above base speed. During acceleration, voltage may be high. (See Figure 27).

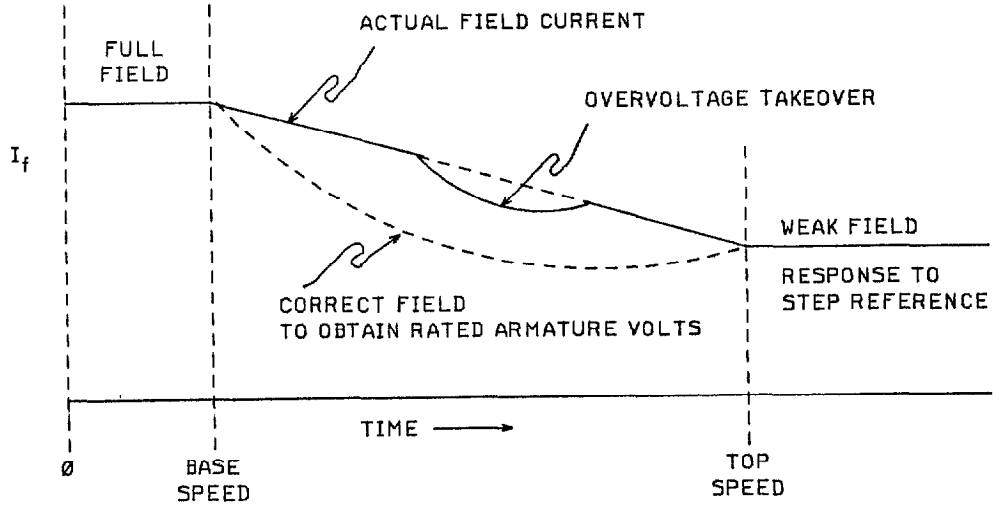


Figure 27

During deceleration, voltage may also be high. (See Figure 28).

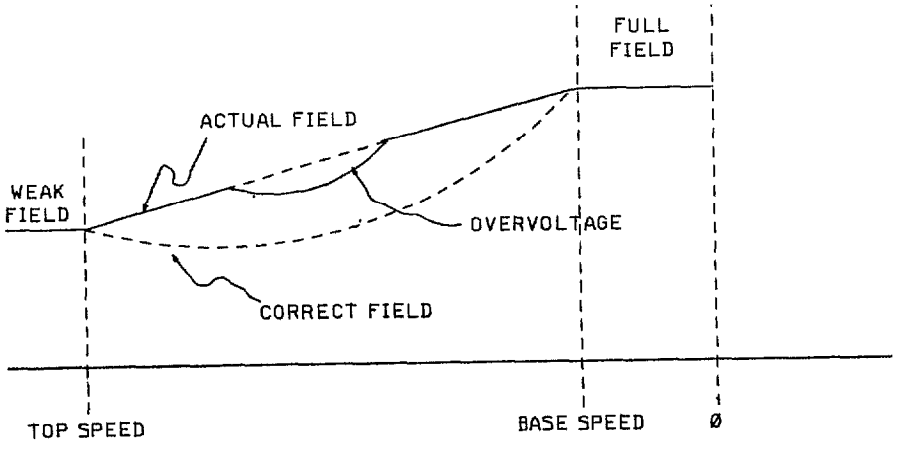


Figure 28

If the overcurrent relay trips, the field may have to be slowed even further. Overvoltage takeover will then control field on rapid changes in reference.

Armature Current Regulator

Under normal conditions, the current regulator should have a response that will reach current limit in .15 to .2 seconds. (See Figure 29).

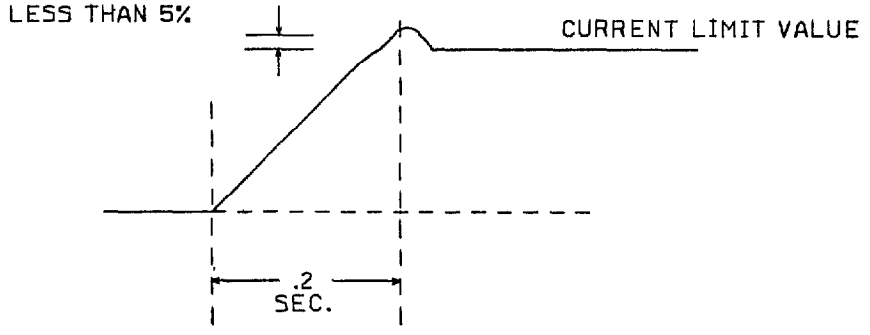


Figure 29

1. Set current limits down to 1 or 2.
2. Turn TSW switch to current.
3. Set RH5 to mid-position (zero current).
4. Push RESET button.

5. Close the loop. (Depress START pushbutton).

6. Check current response with small current steps. Make sure that terminal volts do not exceed rated volts.

7. Check scale stability here. Slowing down current response will improve scale stability.

8. During underspeed control, recheck current response when drive is pushed into current limit.

9. Speed boost. As speed increases, motoring current should decrease, or absorption current should increase. For full rheostat rotation, top speed should produce 100% rated current.

10. Windage. As speed increases, motoring current should decrease, or absorption current should increase. The windage signal is proportional to speed squared.

11. Inertia calibration will be explained later.

If necessary, current limit may be raised during the above tests.

Speed Regulator

Response of the speed regulator should be in accordance with specifications pertaining to this drive. In the absence of specific details, the following is standard.

On 100% impact load, speed droop should not exceed 20% and recovery to 1/2% of operating speed above base speed or 1/2% of base speed below base speed shall be 5 seconds or less.

Response will also determine the amount of scale fluctuation. In general, faster response results in greater scale fluctuation.

1. Turn TSW to speed.

2. Keep current limit set low.

3. Turn RH1 to zero speed.

4. Close loop.

5. Advance speed pot slowly thru the complete speed range to check stability.

6. At a speed just below base speed, put in a small speed step by rapidly twisting the speed knob. The current should not reach current limit.

7. Response should be as shown in Figure 30.

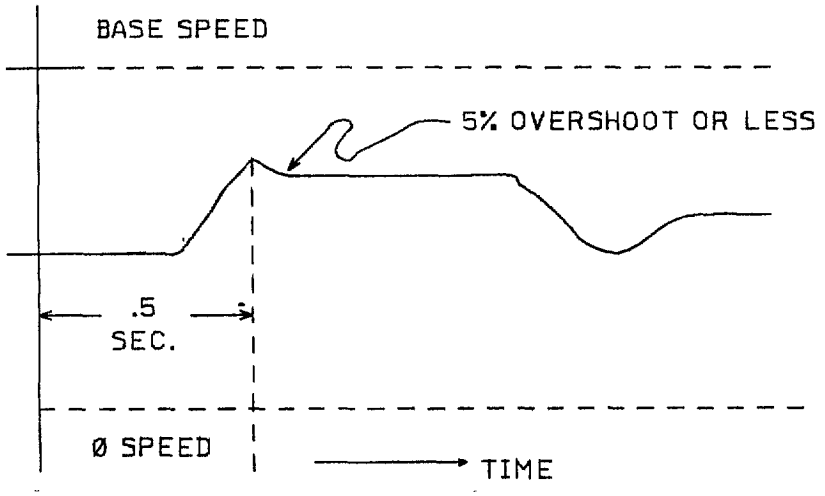
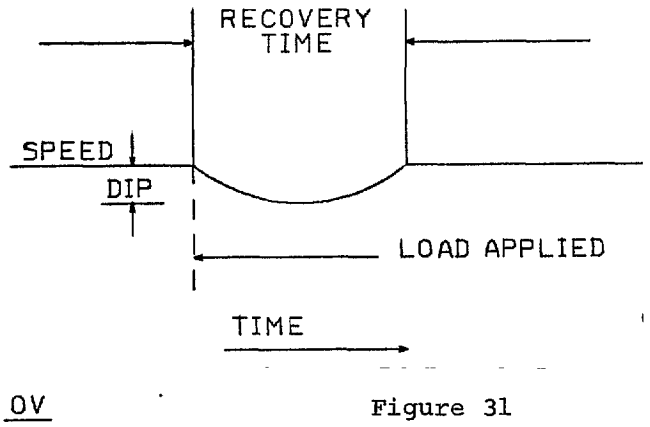


Figure 30

8. Adjust stability if necessary to achieve the response shown in Figure 30 (refer to notes on operational techniques).

9. Turn up current limit.

10. Apply an impact load, if possible. Response should appear as shown in Figure 31.



The numbers to apply to this figure are given at the beginning of this section.

Figure 31

11. From a low value of speed, try emergency stop.

12. Advance speed and try again. Keep repeating until a stop from top speed is made successfully.

13. If reversal of rotation on stop is a problem, a tach lead circuit should be added. (See Figure 32).

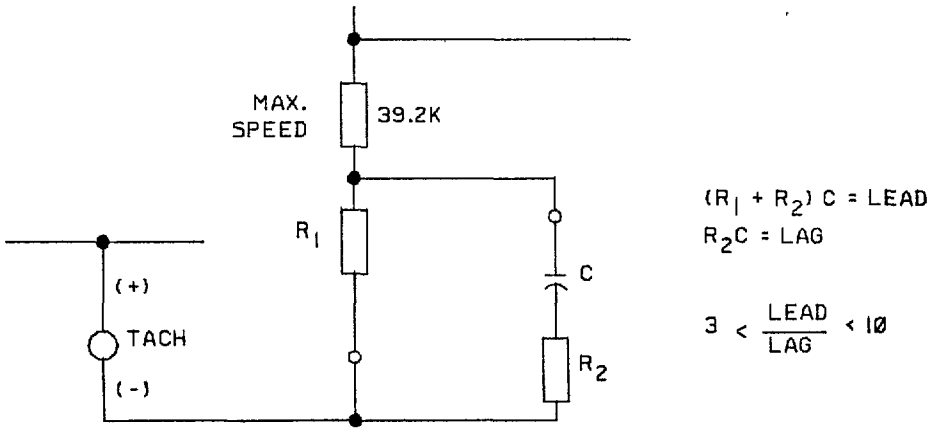


Figure 32

C may vary from .5 to 2.0 MFD.

The stopping characteristic is shown in Figure 33.

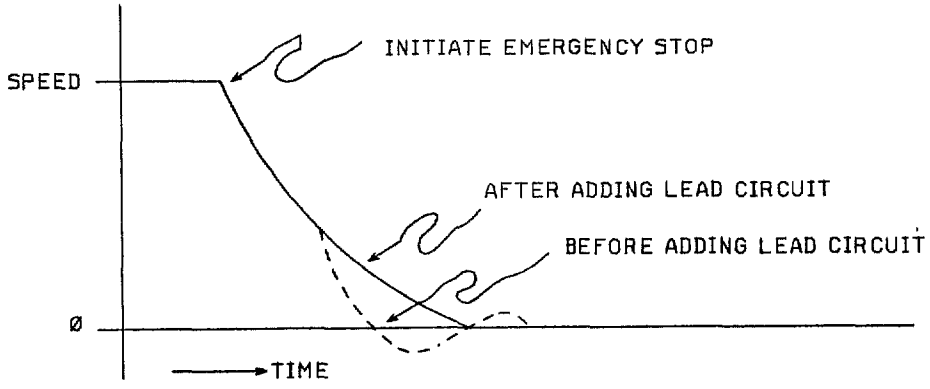


Figure 33

Speed Relay

The speed relay satisfies three functions. Overspeed detection, reverse rotation, and speed match. It should be checked as follows:

1. Speed match: Select speed control. Turn RH1 to zero. Reset EDE.

2. Monitor output of speed amplifier (JR5). It should be zero ±0.2 volt. Turn RH1 until SR picks up. JR5 should be 1.3 volts ± .5 volts. Pickup cannot be adjusted.

3. Overspeed. Switch to current control. Turn RH1 to full rotation (top speed). Let speed increase (uncoupled) by putting in a small amount of motoring current. Carefully nurse speed up to 104% speed (260 volts on tach if tach delivers 250 volts at rated top speed). Adjust 5K pot in speed input card if relay SR does not pick up. (As an alternate: disconnect tach and apply a DC power supply to SR input and adjust as indicated above).

4. Reverse rotation: adjust drive for a small amount of absorption current (uncoupled). As drive goes through zero speed to reverse rotation, SR should pick up below 5% reverse speed. Adjust 1K pot for this value.

Torque Regulator (if present)

Response of the torque regulator should be in accordance with specifications pertaining to this drive. In the absence of specific details, the following is standard (See Figure 34).

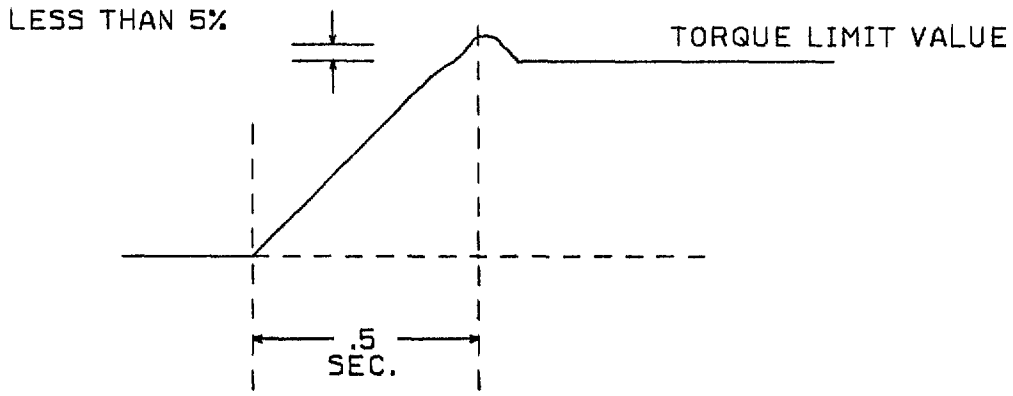


Figure 34

1. Set torque limits down to 1 or 2. The current limit pots should be rotated fully CW.

2. Turn TSW switch to torque.

3. Set RH5 to mid-position (zero torque).

4. Push reset button.

5. Close the loop (depress START pushbutton).

6. Check torque response with small torque steps. Make sure terminal volts do not exceed rated volts.

7. When underspeed control, recheck torque response when drive is pushed into torque limit.

8. Speed boost. As speed increases, motoring torque should decrease, or absorption torque should increase. For full rheostat rotation, top speed should produce 100% rated torque.

9. Windage. As speed increases, motoring torque should decrease, or absorption torque should increase. The windage signal is proportional to speed squared.

10. Inertia calibration will be explained later.

If necessary, torque limit may be raised during the above tests.

Electrical Inertia Calibration

Once current stability has been obtained (as previously described) the electrical inertia reference potentiometer should be calibrated.

System minimum inertia consisting of system mechanical inertia, friction and windage inertia effect and regulator error inertia effect are determined in the following manner:

See sheet 10 and Figure 14 for adjustment of inertia circuit.

System Mechanical Inertia Plus Friction and Windage Inertia Effect

System mechanical inertia plus friction and windage inertia effect are determined by accelerating (and decelerating) the dynamometer and chassis drums (or other connected mechanical devices) over a pre-determined speed range with a given fixed torque. Measure and record the time for acceleration (and deceleration for the given speed range. Adjust the dynamometer field for top speed field current (minimum field current) by having the speed adjusting rheostat turned to the maximum position. Apply constant torque by using a given acceleration (deceleration) torque from a setting on the current adjust rheostat and observe the torque reading on the dynamometer scale. The torque should be kept constant throughout the acceleration (deceleration) period by manually trimming the current adjust rheostat during the test run. The WK^2 is then calculated from the relationship WK^2 = torque times time times 308 divided by change in RPM. ($WR^2 = \frac{T308t}{N}$).

Several runs should be taken and these averaged to obtain the system mechanical inertia plus friction and windage inertia effect. See Figure 35 for typical characteristics.

The deceleration test runs are taken for information purposes only. The difference in system WK^2 between the acceleration and deceleration test runs is of course two times the friction and windage inertia effect.

Regulator Error Inertia effect

In addition to system mechanical inertia and friction and windage inertia effect, we also have a transient regulator error inertia effect which is present in the system as soon as the loop contactor is closed. For normal chassis testing where the dynamometer is used to simulate road load conditions plus system inertia, the transient regulator error results in an absorption torque error when a vehicle is driving the chassis drums and the dynamometer when simulating an acceleration test run. This torque error can be observed by reading the dynamometer scales when the dynamometer is accelerated by a vehicle over a given speed range and while the dynamometer system is attempting to regulate zero road load torque. The order of magnitude of this torque error is a function of the response of the dynamometer system to transient conditions and also a function of the gain of the current regulator. When a vehicle is accelerating the dynamometer under this condition, the torque on the scale is an indication of the transient regulator error torque and does not include the uncradled friction and windage inertia effect of the chassis drums. In essence, the transient regulator error is a load torque imposed on the system and is therefore the sum of the regulator transient error torque plus the uncradled friction and windage torque plus the acceleration torque.

This test should be conducted under closed loop conditions with the dynamometer grade adjusting rheostat adjusted for zero armature current (zero torque) and with the dynamometer field current adjusted for minimum field. Apply an acceleration torque to the dynamometer with the vehicle and accelerate the system from zero to 60 MPH. Record the time for this test run. Such an acceleration test run results in an initial torque peak due to the characteristics of the vehicle. After this transient torque peak, an area of relatively constant torque exists. Measure this system torque and record from the dynamometer weighing system. The regulator transient error inertia effect can now be calculated from this test run as shown in Figure 37.

Total System Minimum Inertia

Total system minimum inertia is therefore the sum of system mechanical inertia, friction and windage inertia effect and regulator transient error inertia. This is the minimum inertia which the total system both mechanically and electrically will impose upon an accelerating vehicle. For an increase in system inertia, simulated position electrical inertia can be added. For a decrease in system inertia, simulated negative electrical inertia can be added.

Calibration of Electrical Inertia Potentiometer (Positive)

An accurate calibration of the dial position of the electrical inertia potentiometer is required to be able to determine the total system inertia which is being represented at any one time. Conduct the calibration in the following manner:

Accelerate the dynamometer over a given speed range for zero dial position of the inertia potentiometer. Measure and record the speed interval and time for this test run. During the test run, adjust the dynamometer field current for minimum field (top speed condition) and apply a motoring accelerating torque by presetting the given torque with the dynamometer grade adjusting potentiometer. During this acceleration run, the grade adjusting potentiometer should not be disturbed. With zero setting on the inertia potentiometer dial, the system inertia would consist of the system mechanical inertia plus the friction and windage inertia and the regulator transient error inertia. Three or four test runs should be taken and the average time over the same speed interval obtained. The system inertia for this setting of zero inertia potentiometer dial has been previously determined and is now known.

Adjust the inertia potentiometer dial for the one (1) position setting and conduct a similar acceleration test run. Record the time over the same speed range and with the same acceleration torque as for the zero dial setting run. Several test runs at this inertia potentiometer setting should be made and the average time obtained. The ratio of the times for a given dial position run and the zero dial position run multiplied by system minimum inertia is now the new total system inertia at that given inertia potentiometer dial setting:

$$\frac{t1}{t0} \times (WK^2)\ 0 = (WK^2)1$$

These tests will then continue throughout the range of zero-10 on the inertia potentiometer dial.

Calibration of Electrical Inertia Potentiometer (Negative)

The procedure for the calibration of the electrical inertia potentiometer for negative electrical inertia is identical to that used for positive electrical inertia.

An inertia potentiometer calibration curve, which graphically depicts these results, is shown in Figure 38.

It is apparent from this calibration curve that the range of positive and negative electrical inertia are not identical. The range of positive electrical inertia to system minimum inertia is normally 2 to 1 or more. The ratio for negative electrical inertia is normally 0.5 to 1. See the requisition for exact ranges required.

The range of electrical inertia, both positive and negative, are determined by the capacitor in the inertia reference circuit.

START		STOP		SCALE			AVG
MPH	RPM	MPH	RPM	RDG	TIME	WK^2	WK^2
20	140	110	770	150	75.8	11100	
20	140	110	770	150	75.3	11050	
20	140	110	770	150	74.9	10950	11030
20	140	110	770	150	75.0	11000	
20	140	110	770	150	75.3	11050	
20	140	110	770	300	37.8	11100	
20	140	110	770	300	37.7	11100	
20	140	110	770	300	37.6	11050	11087
20	140	110	770	300	37.8	11100	

$$WK^2 = \frac{T \times t \times 308}{\Delta N}$$

WK^2 = Lb. Ft.^{2}

T = torque = Lb. Ft. = Scale RDG X 2

t = Time = Seconds

Δ N = Speed Change = RPM = 630

Figure 35
System Mechanical Inertia Plus Friction and Windage Inertia

SCALE RDG	TORQUE	AVG TORQUE	MPH	TIME
12	24		32	
11.7	23.4		35	
11.5	23		36	
11.0	22		40	
10.0	20	20.4	45	
9.2	18.4		50	
9.0	18		52	
9.5	19.6		53	0 - 60
9.8	19.6		55	MPH
9.5	19		58	
9.2	18.4		60	

Regulator Error Inertia Plus Friction and Windage Inertia = $\frac{T \times t \times 308}{\Delta N}$

$\frac{20.4 \times 44 \times 308}{420}$

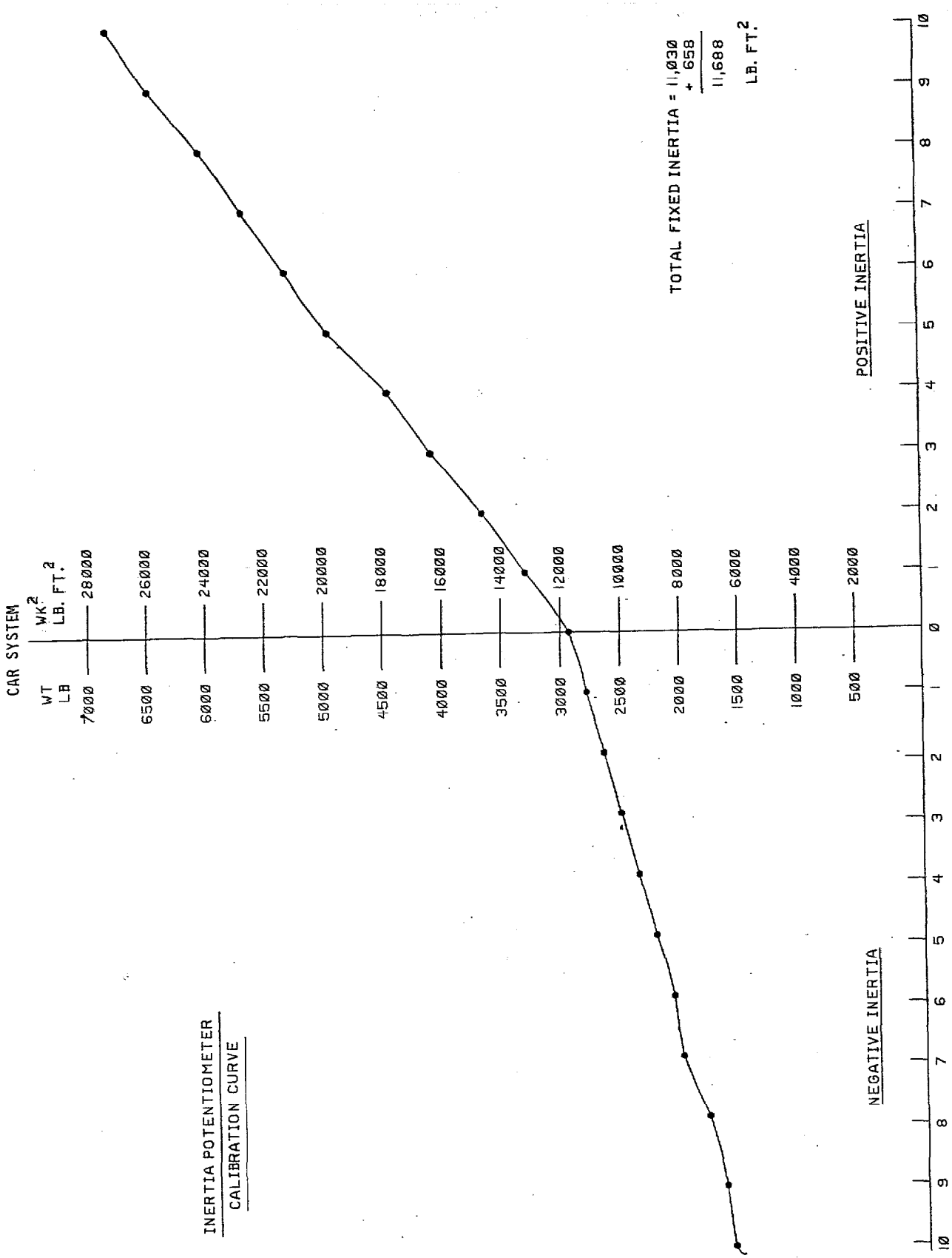
$= 685 \text{ Lb. Ft.}^2$

T = 20.4 Lb. Ft.

ΔN = 420 RPM

t = 44 Seconds

Figure 36
Regulator Error Inertia Plus Friction and Windage Inertia


INERTIA POTENTIOMETER
CALIBRATION CURVE
CAR SYSTEM
WT LB
WK² LB. FT.²
7000 28000
6500 26000
6000 24000
5500 22000
5000 20000
4500 18000
4000 16000
3500 14000
3000 12000
2500 10000
2000 8000
1500 6000
1000 4000
500 2000
TOTAL FIXED INERTIA = 11,030 + 658
11,688 LB. FT.²
POSITIVE INERTIA
NEGATIVE INERTIA
0 1 2 3 4 5 6 7 8 9 10

OPERATING INSTRUCTIONS

The sequence of operation is outlined below:

1. Turn the reversing switch to either FORWARD or REVERSE. This, of course, determines the direction of rotation of the dynamometer, and energizes the control circuit.

2. Press the motor-generator set START pushbutton. The motor-generator set will start, and will automatically sequence the dynamometer system auxiliaries to start, including the blower motors, pump motors, and SCR exciter. Each of these auxiliaries is protected with an overload relay or fuse. The motor-generator set STOP pushbutton will stop all of these auxiliaries and exciters as well as the motor-generator set.

3. Turn the transfer switch to SPEED CONTROL. It is more desirable to start up under SPEED CONTROL than under TORQUE CONTROL or CURRENT CONTROL.

4. Turn the regulation potentiometer to MAXIMUM and the speed control rheostat to MINIMUM.

5. Push the dynamometer START pushbutton, thus closing the armature loop circuit contactor.

6. The speed control rheostat can now be advanced to obtain the desired dynamometer speed.

7. During any changes, the operator should watch the RPM indicator, voltmeter, and ammeter in order to check operation of the system.

8. The test engine can now be energized and automatically loaded.

9. If it is desired to protect the equipment under test from too high a torque while the dynamometer is motoring, the motor current limit potentiometer may be adjusted counterclockwise. This will reduce the reference of the motoring current limit circuit, and thus permit the current limit circuit to take over at lower value of current.

10. For normal stopping of the system, the load should be removed from the dynamometer and the speed reduced to zero by means of the speed control potentiometer. The dynamometer STOP pushbutton can then be pressed to open the loop contactor.

11. For emergency stopping, press the emergency stop pushbutton. This will rapidly bring the drive to rest by regenerative braking. The RESET pushbutton must then be depressed before the drive can be started.

12. Indicating lights tell the operator when the motor-generator set is running or stopped, and when the loop contactor is open or closed. The loop contactor OPEN light will not come on until all protective devices are in their normal operating condition. Furthermore, some systems are supplied with additional indicating lights or annunciator drops to indicate that a protective device (described above) has operated.

MAINTENANCE

All electrical apparatus will provide maximum trouble-free service if given the benefit of preventative maintenance inspections and periodic cleanings. Although the frequency of the inspection periods will depend on operating conditions, it is important that a definite inspection routine and inspection schedule be followed. Again, maintenance instructions in the individual device instructions are most helpful.

Renewal Parts

Order renewal parts by part number and description, and identify the equipment for which they are required. Structural parts and complete operating devices are identified on the equipment drawing list. They are available on request.

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